

GROUNDCONTROL
from Integer Audio

Nodal 30

Sub-60 Hz Bass Modal Treatment

For professional monitoring, recording, and critical listening environments

“Simulate your room. Predict the result. Then verify it with measurements.”

THE PROBLEM WITH BASS IN SMALL ROOMS

Most monitoring problems in small rooms are not caused by the speakers. They are caused by room modes.

When low-frequency sound reflects between room boundaries, standing waves form at specific frequencies. At these frequencies, acoustic energy accumulates and decays slowly, lingering long after the signal has stopped. The result is uneven bass response throughout the room and low-frequency notes that persist longer than intended.

In practical terms:

- Kick drums sound bloated rather than tight and punchy
 - Bass notes disappear in some listening positions and boom in others
 - Low-frequency balance changes as you move around the room
 - Mixes that sound correct in the studio fail to translate to other playback systems
- **EQ changes level. It does not change decay time. A frequency that rings for two seconds will still ring for two seconds after EQ is applied.**

Target decay time (T60) for professional monitoring: 0.25–0.55 seconds

T60 Decay Time	What You Hear	Mix Impact
Above 1.5 s	Heavy, one-note bass. Kick and bass merge.	Bass adjustments don't translate. Every mix is too thin elsewhere.
0.75–1.5 s	Colored, uneven. Some notes louder than others.	Low end unreliable. Decisions made in the room don't hold.
0.25–0.55 s ✓	Controlled, even, articulate.	Bass decisions translate. The room works for you.

Table 1 — Bass decay time and its effect on monitoring quality

WHY LOW-FREQUENCY TREATMENT IS DIFFICULT

Traditional broadband absorbers work by dissipating acoustic energy as air moves through porous materials such as fiberglass or mineral wool. They are highly effective at mid and high frequencies and can provide useful absorption down to approximately 60 Hz when installed with sufficient depth. Below 40 Hz, wavelengths exceed 8 meters (26 feet). Achieving significant absorption at these frequencies using porous materials alone requires impractically thick treatments. As a result, rooms that appear heavily treated often still exhibit excessive low-frequency decay and modal ringing in the deep bass region.

→ The Nodal 30 was engineered specifically to reduce deep-bass modal decay where conventional broadband absorbers become impractical.

HOW THE NODAL 30 WORKS

The Nodal 30 is a Tuned Diaphragmatic Absorber (TDA).

It uses a precision-tuned membrane and enclosed air cavity to absorb low-frequency energy. When bass energy near the tuning frequency reaches the unit, the membrane vibrates in a controlled manner, converting acoustic energy into heat. This enables effective absorption at frequencies where conventional porous treatments become impractically large.

The tuning frequency is determined by the membrane mass and cavity depth. The Quality factor represents the impedance-matched operating point, where energy transfer from the room to the absorber is maximized. This configuration provides the greatest integrated absorption across the intended operating range.

Tuning Frequency	35 Hz
Quality Factor (Q)	1.0 — impedance-matched for maximum absorption
Operating Range (–3 dB)	Approximately 18–52 Hz
Membrane	Richlite + Mass Loaded Vinyl
Cavity Depth	215 mm
Membrane Area	24" × 30" (610 × 762 mm)

THE COMPLETE TREATMENT SYSTEM

The Nodal 30 is designed to complement conventional broadband absorption. The two approaches address adjacent low-frequency ranges with only modest overlap, providing more complete control than either treatment can achieve alone.

Nodal 30 and broadband panels perform different jobs.

NODAL 30

20 – 60 Hz

Controls deep-bass room modes
Effective where porous absorbers become impractically thick
Designed specifically for modal decay reduction

BROADBAND PANELS

60 Hz and above

Controls upper bass, midrange, and high frequencies
Reduces reflections and overall reverberation
Essential for balanced room treatment

Neither is a substitute for the other.

Together, they provide continuous control from the deepest room modes through the remainder of the audible spectrum.

PLACEMENT

The Nodal 30 is designed for corner placement to achieve maximum performance.

Room-mode pressure is highest where room boundaries intersect. In a rectangular room, corners represent the highest-pressure regions for axial modes. Placing an absorber in a corner can increase its effective modal absorption by up to four times compared to a mid-wall location. Multiple units may be stacked to further increase low-frequency absorption and extend modal control.

Recommended placement priority:

- Floor corners (primary location, recommended for all installations)
- Ceiling corners (similarly effective to floor corners; ideal when floor space is limited)
- Wall-wall corners at mid-height (good modal coupling; use when floor and ceiling corners are unavailable)
- Mid-wall positions (reduced modal coupling; best used as a supplemental treatment location)

HOW MANY UNITS DO I NEED?

The answer depends on your room dimensions, construction, and target performance. Use the GroundControl Simulation Tool for a room-specific recommendation.

Configuration	Typical Outcome	Best For
4 units (1 per corner)	T60 at sub-40 Hz modes: 1.2–1.5 s → 0.38–0.50 s. Consistent improvement across all axial modes.	Standard starting configuration for most project and professional studios.
6–8 units (2 per corner)	Deeper treatment. Approaches saturation limit at most mode frequencies. Projections shown in the simulation.	Mastering rooms and highly optimized monitoring environments.

Table 2 — Indicative configurations. Use the GroundControl Simulation for room-specific predictions.

THE GROUNDCONTROL SIMULATION TOOL

Every Nodal 30 installation is backed by GroundControl Acoustic Modal Simulation, a physics-based modeling tool that predicts low-frequency room performance and treatment effectiveness before you purchase or install a single panel.

What it does:

- Calculates your room’s axial mode frequencies from its dimensions
- Predicts modal decay time (T60M) at each mode before and after treatment
- Models the contribution of broadband absorption based on coverage area and panel depth
- Determines the recommended number of Nodal 30 units required to achieve your target performance
- Generates before-and-after performance predictions for any treatment configuration

Calibrated to your room using REW:

Import a REW waterfall measurement from your untreated room. The GroundControl simulation calculates T60M at each modal frequency and calibrates the simulation to your room's actual acoustic behavior. After installation, import a second measurement to verify measured performance against the model's predictions.

“We simulate your room. Predict the result. Then verify it with measurements.”

EXPECTED RESULTS

The objective of the Nodal 30 is simple:

Reduce excessive low-frequency modal decay time toward the 0.25–0.55 second target range used in professional monitoring environments.

As modal ringing is reduced, bass becomes tighter and more articulate, pitch definition improves, and mix decisions become more reliable and consistent across playback systems.

COMMON QUESTIONS

Can EQ replace bass treatment?

No. EQ changes level, but it does not reduce decay time. A room mode that takes two seconds to decay will continue to ring for two seconds regardless of EQ settings. EQ can reduce the amplitude of a resonance, but it cannot remove the underlying acoustic problem.

Do I need broadband panels as well?

For a complete monitoring environment, yes. Broadband panels address frequencies above approximately 60 Hz, while the Nodal 30 is designed for deep-bass modal control below 60 Hz. Together they provide continuous control from the lowest room modes through the remainder of the audible spectrum. Neither is a substitute for the other.

How many units do I need?

The GroundControl Acoustic Modal Simulation determines the recommended quantity for your specific room. Most rooms in the 4.5–7 m (15–23 ft) size range achieve the target decay-time range with four units (one per corner). Larger rooms may benefit from additional units.

Why isn't ISO 354 data provided?

ISO 354 measurements are performed in a diffuse-field reverberation chamber above the Schroeder frequency. The Nodal 30 is designed to operate below this region, where room behavior is dominated by discrete modes rather than diffuse sound fields.

For low-frequency modal treatment, in-room decay measurements provide a more meaningful indication of performance. GroundControl uses REW waterfall data to calculate modal decay times (T60M) before and after treatment, allowing performance to be predicted and verified in the actual listening environment.

How long should I wait before measuring?

Allow 24 hours after installation before performing verification measurements. This allows the Richlite membrane and Mass Loaded Vinyl (MLV) layer to fully acclimate to room temperature and humidity, ensuring stable and repeatable performance.

SPECIFICATIONS

Tuning Frequency	35 Hz
Quality Factor	Q = 1.0 (impedance-matched)
Membrane Area	24" × 30" (610 × 762 mm)
Active Cavity Depth	215 mm
Membrane Construction	Richlite 1/4" + Mass Loaded Vinyl
Internal Damping	OC705 compressed fiberglass
Recommended Placement	Room corners — floor or ceiling primary
−3 dB Coverage	Approximately 20–62 Hz
Simulation Support	GroundControl Acoustic Modal Simulation (r61) · REW waterfall T60M calibration

Performance predictions should be validated using REW waterfall T60M measurements before and after installation. Predictions are indicative; actual results depend on room construction, unit placement, and interaction with other treatment in the room.